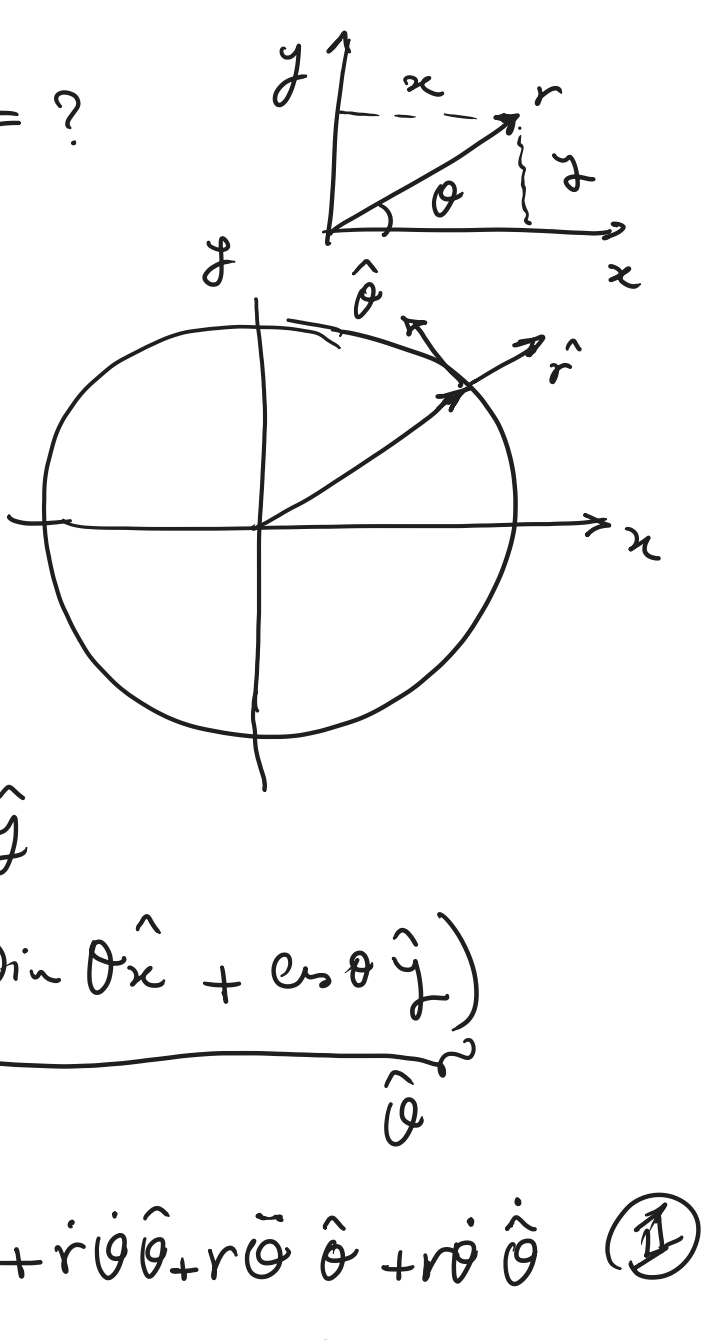
Problem 1) Goal: Express eqn of motion $\vec{F}=m\vec{a}$ in 2D Polar coordinates.

$$\vec{F}=F_r\hat{r}+F_\theta\hat{\theta}\ ;\ F_r=?\ ;\ F_\theta=?$$



We have $\vec{F}=m\vec{a}=m\ddot{\vec{r}}$, so our goal is to write $\ddot{\vec{r}}$ in terms of $\hat{r}$ and $\hat{\theta}$.

$$\vec{r}=r\hat{r}$$

But $\hat{r}$ and $\hat{\theta}$ are time-dependent, unlike $\hat{x},\hat{y}$.

$$\dot{\vec{r}}=\dot{r}\hat{r}+r\dot{\hat{r}}\ ;\ \hat{r}=\cos\theta\hat{x}+\sin\theta\hat{y}$$

$$\Rightarrow \dot{\hat{r}}=-\hat{x}\dot{\theta}\sin\theta+\hat{y}\dot{\theta}\cos\theta=\dot{\theta}\underbrace{(-\sin\theta\hat{x}+\cos\theta\hat{y})}_{\hat{\theta}}$$

$$\Rightarrow \dot{\hat{r}}=\dot{\theta}\hat{\theta}\quad \text{Ⓘ}$$

$$\Rightarrow \dot{\vec{r}}=\dot{r}\hat{r}+r\dot{\theta}\hat{\theta}\Rightarrow \ddot{\vec{r}}=\ddot{r}\hat{r}+\dot{r}\dot{\hat{r}}+\dot{r}\dot{\theta}\hat{\theta}+r\ddot{\theta}\hat{\theta}+r\dot{\theta}\dot{\hat{\theta}}\quad \text{(II)}$$

$$\hat{\theta}=-\sin\theta\hat{x}+\cos\theta\hat{y}\Rightarrow\dot{\hat{\theta}}=-\dot{\theta}\cos\theta\hat{x}-\dot{\theta}\sin\theta\hat{y}=-\dot{\theta}\hat{r}\quad\text{(III)}$$

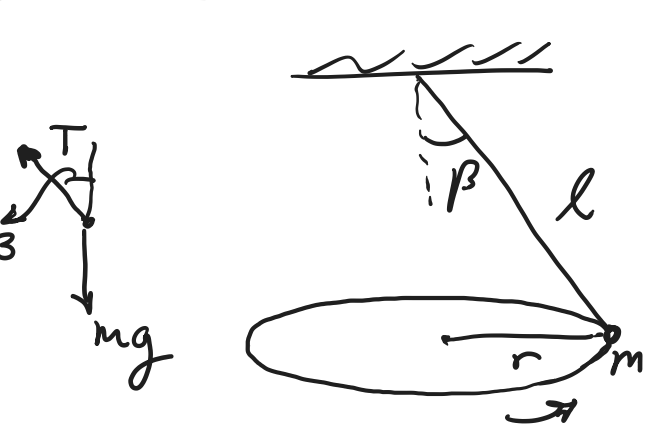
$$\text{I, II, III}\Rightarrow\ddot{\vec{r}}=\ddot{r}\hat{r}+\dot{r}(\dot{\theta}\hat{\theta})+r\dot{\theta}\hat{\theta}-r\dot{\theta}^2\hat{r}$$
$$=(\ddot{r}-r\dot{\theta}^2)\hat{r}+(r\ddot{\theta}+2\dot{r}\dot{\theta})\hat{\theta}$$

$$\vec{F}=m\ddot{\vec{r}}=F_r\hat{r}+F_\theta\hat{\theta}\Rightarrow\begin{cases}F_r=m(\ddot{r}-r\dot{\theta}^2)\\F_\theta=m(r\ddot{\theta}+2\dot{r}\dot{\theta})\end{cases}\ \checkmark$$

Remarks:

① The term $m\ddot{r}$ is intuitive: $F=ma$ in the radial direction.
② $mr\ddot{\theta}$ is also similarly intuitive.
③ $-mr\dot{\theta}^2=-mv^2/r$ is the centripetal force.
④ $2m\dot{r}\dot{\theta}$ is associate with Coriolis force.
Its presence is to ensure conservation of angular momentum.

Problem 2) known: β, m, l, unknown: ω



$$F=ma\ ;\ F_r=T\cos\beta-mg=0\Rightarrow T=\frac{mg}{\cos\beta}$$

From problem ①, $F_r=m(\ddot{r}-r\dot{\theta}^2)$
$$=0-mr\dot{\theta}^2=T\sin\beta=mg\tan\beta$$

$$r=l\sin\beta\Rightarrow\omega^2=\frac{T\sin\beta}{ml\sin\beta}\Rightarrow\omega=\sqrt{\frac{g}{l\cos\beta}}$$

Remarks:

$$\lim_{\beta\to 90}\omega=\infty\ ;\ \lim_{\beta\to 0}\omega=\sqrt{g/l}$$

Problem 3) knowns: R: radius of parking lot   v: car's constant speed.
θ: inclination angle of lot   μ: static friction coefficient.

a) The centripetal force required for keeping the car in circle is, $\vec{F}_c=-\frac{mv^2}{R}\hat{r}$, represented by the inward components of gravity and friction forces: $F_{c,min}=\frac{mv_{min}^2}{R}\leqslant\mu N-mg\sin\theta=mg(\mu\cos\theta-\sin\theta)$

$$\Rightarrow\boxed{v_{min}\leqslant\sqrt{gR(\mu\cos\theta-\sin\theta)}}$$

(b) The radial velocity is constant, thus, all the tangential forces must cancel out on the sides: $mg\sin\theta=F_{f,tangential}$ and the inward friction must be $F_{f,inward}\geqslant F_c=\frac{mv_{max}^2}{2}$; $F_f=F_{f,tan}^2+F_{f,in}^2=\mu N=\mu mg\cos\theta$

$$\Rightarrow(mg\sin\theta)^2+\left(\frac{mv_{max}^2}{2}\right)^2\leqslant(\mu mg\cos\theta)^2\Rightarrow\boxed{v_{max}\leqslant\sqrt{gR}\left(\mu^2\cos^2\theta-\sin^2\theta\right)^{1/4}}$$

Problem 4) Recall from Problem 1 above that the tangential component of force in polar coordinates is: $\vec{F}_\theta=m(r\ddot{\theta}+2\dot{r}\dot{\theta})\hat{\theta}=m\dot{r}\dot{\theta}\hat{\theta}$ which means $\ddot{\theta}$ must be zero for the equality to hold:

$$r\ddot{\theta}=0\Rightarrow\ddot{\theta}=0\Rightarrow\dot{\theta}=C=\text{const.}\Rightarrow\theta=Ct+D$$

The corresponding radial component of polar force is: $\vec{F}_r=(\ddot{r}-r\dot{\theta}^2)\hat{r}$ which is zero in this problem. $\Rightarrow\ddot{r}=r\dot{\theta}^2=rC^2$ When the derivative of a function is proportional to itself, it is reminiscent of exponential function. I this case, $r(t)=ae^{ct}+b^{-ct}$ is a solution to $\ddot{r}=rc^2$. Noting that $ct=\theta-D\Rightarrow r(\theta)=(ae^{-D})e^{\theta}+(be^{+D})e^{-\theta}=\boxed{Ae^{\theta}+Be^{-\theta}}$

Problem 5) $F=F(x)\ ;\ F=m\frac{dv}{dt}=m\frac{dv}{dx}\frac{dx}{dt}=mv\frac{dv}{dx}$

$$\Rightarrow F(x)dx=mv\,dv\Rightarrow\int_{x_0}^{x}F(x')dx'=\frac{1}{2}mv'^2\Big|_{v_0}^{v}$$

$$\Rightarrow\boxed{\frac{1}{2}mv_0^2=\frac{1}{2}mv^2-\int_{x_0}^{x}F(x')dx'=\text{Const.}}$$

Let $U(x)-U(x_0)=-\int_{x_0}^{x}F(x')dx'$

$\Rightarrow F(x)=-\frac{dU(x)}{dx}$ Note that potential energy is defined in terms of a reference point, always.

Problem 6) known: M, m, r $F_r=\frac{GMm}{r^2}$

$$U(r)-U(r_0)=-\int_{r_0}^{r}\frac{GMm}{r'^2}dr'=-\frac{GMm}{r}+\frac{GMm}{r_0}$$

$$r_0\to\infty\Rightarrow U(r_0)\to 0\Rightarrow U(r)=-\frac{GMm}{r}$$

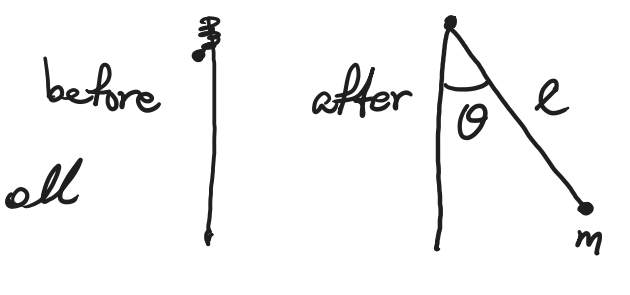
Problem 7) known: m, y (height), R (Earth radius); $y\ll R$

$$U(R+y)-U(R)=-\frac{GMm}{R+y}+\frac{GMm}{R}=-\frac{GMm}{R}\left(\frac{1}{1+y/R}-1\right)$$

$$y\ll R\Rightarrow\frac{y}{R}\to 0\Rightarrow\frac{1}{1+y/R}=1-\frac{y}{R}+(y/R)^2-O^3(y/R)$$
$$\approx 1-y/R$$

$$\Rightarrow\boxed{\Delta U=\frac{GMm\,y}{R^2}=gmy}$$

Problem 8) known: l: string length; frictionless poll; massless string; constant (slow) unwinding vertical speed of mass m.
unknown: the angle between pole and string when completely unwound. θ=?



When completely unwound, m is revolving around the poll with the centripetal force $F_c=\frac{mv^2}{r}=\frac{mv^2}{l\sin\theta}=T\sin\theta$ Ⓘ

For the vertical direction, $\Sigma F_v=0\Rightarrow mg=T\cos\theta$ (II)

$$\text{Ⓘ, (II)}\Rightarrow mv^2=lT\sin^2\theta=mgl\sin\theta\tan\theta\quad\text{(III)}$$

However, we know from conservation of energy that: $\frac{1}{2}mv^2=mgh=mgl\cos\theta$ (IV)

$$\text{(III), (IV)}\Rightarrow 2mgl\cos\theta=mgl\sin\theta\tan\theta\Rightarrow\tan^2\theta=2\Rightarrow\boxed{\theta=\arctan(\sqrt{2})}$$

Problem 9) The work-energy theorem of lecture ocw6 states that

$$\frac{1}{2}m(v^2-v_0)=\int_{x_0}^{x}F(x')dx'=0\quad\text{because } v=v_0$$

Thus $\int_{x_0}^{x}F(x')dx'=0$. This makes sense because the sum of the work done by the force of gravity on the book and the upward normal force of the same magnitude but opposite direction is exactly zero: $N=+mg\hat{y}$

$$\Rightarrow\Sigma W=\int_{y_0}^{y}(N-mg)dy'=\int 0\,dy=0\ \checkmark$$

Problem 10) In 1D, we have, $F_x=mv_x\frac{dv_x}{dx}$. By analogy,

$$\vec{F}_x+\vec{F}_y+\vec{F}_z=m\left(v_x\frac{dv}{dx}\hat{x}+v_y\frac{dv_y}{dy}\hat{y}+v_z\frac{dv_z}{dz}\hat{z}\right)=$$

$$\Rightarrow F_xdx+F_ydy+F_zdz=m(v_xdv_x+v_ydv_y+v_zdv_z)$$

$$\Rightarrow\vec{F}.d\vec{r}=m\vec{v}.d\vec{v}\Rightarrow\underbrace{\int\vec{F}.d\vec{r}}_{\substack{-U\\\text{potential energy}}}=\underbrace{\frac{1}{2}mv^2}_{\substack{\text{kinetic energy}\\T}}-\underbrace{\frac{1}{2}v_0^2}_{\substack{\text{initial kinetic energy}\\E}}$$

$$\Rightarrow\boxed{T+U=E=\text{constant.}}$$

Problem 11) Recall the definition of conservative force and its relation to potential energy: $dU(\vec{r})=-\vec{F}(\vec{r}).d\vec{r}=-(F_xdx\hat{x}+F_ydy\hat{y}+F_zdz\hat{z})$

$$=\frac{\partial U(\vec{r})}{\partial x}dx+\frac{\partial U(\vec{r})}{\partial y}dy+\frac{\partial U(\vec{r})}{\partial z}dz$$

$$\Rightarrow\begin{cases}F_x=-\frac{\partial U}{\partial x}\\F_y=-\frac{\partial U}{\partial y}\\F_z=-\frac{\partial U}{\partial z}\end{cases}\Rightarrow\boxed{\vec{F}(r)=-\nabla U(\vec{r})}$$

But we know already from calculus that curl of gradient of a function is zero: $\vec{\nabla}\times\vec{\nabla}U=0\Rightarrow\nabla\times(-F(r))=0\Rightarrow\boxed{\vec{\nabla}\times\vec{F}(\vec{r})=0}$

An interesting observation: the Stokes Theorem states that the line integral of a vector field over a loop equals the surface integral of its curl over the corresponding enclosed surface. This implies that the potential $U(r)=-\int_{r_0}^{r_1}\vec{F}(\vec{r}).d\vec{r}$ is independent of the path taken from r_0 to r_1 because any two path segments from r_0 to r_1 back to r_0 forms a closed loop.

Problem 12) The definition of curl in Spherical coordinates is given by,

$$\nabla\times\vec{A}=\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(A_\varphi\sin\theta)-\frac{\partial A_\varphi}{\partial\varphi}\right)\hat{r}$$
$$+\frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\varphi}-\frac{\partial}{\partial r}(rA_\varphi)\right)\hat{\theta}$$
$$+\frac{1}{r}\left(\frac{\partial}{\partial r}(rA_\theta)-\frac{\partial A_r}{\partial\theta}\right)\hat{\varphi}$$

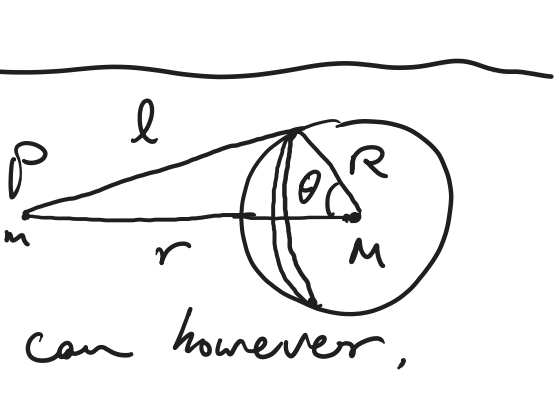
$$\vec{A}=\vec{r}=r\hat{r}\Rightarrow A_\theta=A_\varphi=0\ ;\ A_r=r\Rightarrow\frac{\partial A_r}{\partial\theta}=0\ ;\ \frac{\partial A_\theta}{\partial r}=0;\ \frac{\partial A_r}{\partial\varphi}=0$$

$\Rightarrow\boxed{\vec{\nabla}\times\vec{r}=0}$ We can also prove this using the more involved Cartesian coordinates.

$$\vec{\nabla}\times\vec{r}=\left(\frac{\partial r_z}{\partial y}-\frac{\partial r_y}{\partial z}\right)\hat{x}+\left(\frac{\partial r_x}{\partial z}-\frac{\partial r_z}{\partial x}\right)\hat{y}+\left(\frac{\partial r_y}{\partial x}-\frac{\partial r_x}{\partial y}\right)\hat{z}$$

$$\vec{r}=\frac{x}{r}\hat{x}+\frac{y}{r}\hat{y}+\frac{z}{r}\hat{z}\Rightarrow\left\{\begin{aligned}\frac{\partial r_z}{\partial y}-\frac{\partial r_y}{\partial z}&=0-0=0\\\frac{\partial r_x}{\partial z}-\frac{\partial r_z}{\partial x}&=0-0=0\\\frac{\partial r_y}{\partial x}-\frac{\partial r_x}{\partial y}&=0-0=0\end{aligned}\right\}\Rightarrow\boxed{\vec{\nabla}\times\vec{r}=0}$$

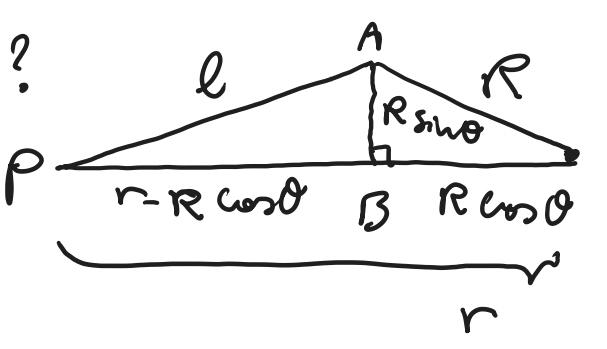
Problem 13) Obtaning the force by integrating the point forces over all elements of sphere would be daunting because it is a vector integration. We can however, achieve the goal by computing the potential energy of the point mass m with respect to the sphere M and the $F(r)=-\nabla U(\vec{r})$



To compute the potential, consider a slice of the sphere making an angle θ with the sphere center. Let dM represent the resulting ring's mass, and its area is given by $dA=2R(\sin\theta)\pi R d\theta$. Also, let $\rho=\frac{M}{4\pi R^2}$ be the mass density of the sphere.

$$\text{Thus, } dM=\rho dA=\frac{2\pi MR^2(\sin\theta)d\theta}{4\pi R^2}=\frac{M}{2}\sin\theta\,d\theta$$

Because, dM is distributed uniformly at the same distance from p, it follows that, $dU(l)=-\frac{GmdM}{l}$ what is l?



$$\Rightarrow l=\sqrt{r^2+R^2-2rR\cos\theta}$$

$$\Rightarrow dU(l)=-\frac{GmM\sin\theta d\theta}{2\sqrt{r^2+R^2-2rR\cos\theta}}\Rightarrow U(r)=-\frac{GmM}{2}\int_0^{\pi}\frac{\sin\theta d\theta}{\sqrt{r^2+R^2-2rR\cos\theta}}$$

$$\Rightarrow U(r)=-\frac{GMm}{2rR}\sqrt{r^2+R^2-2rR\cos\theta}\Big|_0^{\pi}$$

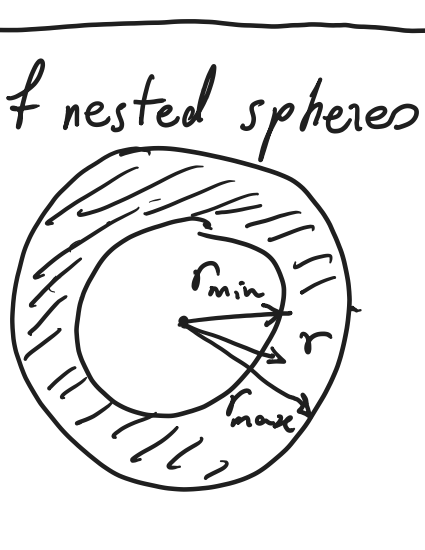
$$U(r)=-\frac{GMm}{rR}\left((R+r)-(r-R)\right)=-\frac{GMm\,2R}{2rR}=\boxed{-\frac{GMm}{r}}\quad r\geqslant R$$

This means that a uniform sphere gravitationally behaves similar to a point mass.

If $r\leqslant R$, then $U(r)=-\frac{GMm}{rR}\left((R+r)-(R-r)\right)=\boxed{-\frac{GMm}{R}\quad r\leqslant R}$

$$\Rightarrow\begin{cases}F=-\frac{GMm}{r^2}\hat{r} & r\geqslant R\quad\text{as if sphere is point like.}\\F=0 & r<R\quad\text{no gravity felt inside sphere.}\end{cases}$$

Problem 14) We can consider the thin shell as a collection of nested spheres with different radii. Thus,



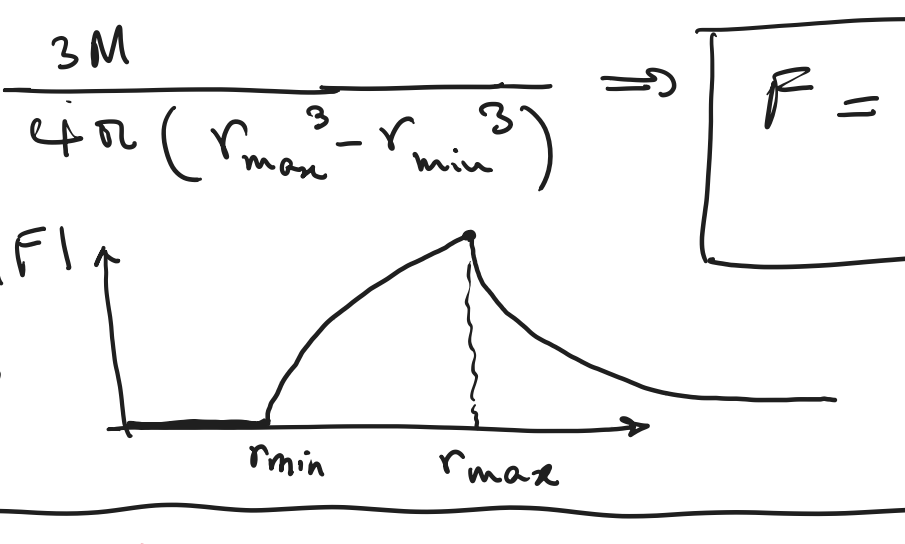
$$r\geqslant r_{max}\ :\ F=-\frac{GMm}{r^2}$$

$$r\leqslant r_{min}\ :\ F=0$$

$$r_{min}\leqslant r\leqslant r_{max}\ :\ F=-\frac{GmM(r)}{r^2}=-\frac{Gm\rho}{r^2}\frac{4\pi}{3}(r^3-r_{min}^3)$$

$$\rho=\frac{3M}{4\pi(r_{max}^3-r_{min}^3)}\Rightarrow\boxed{F=-\frac{GMm}{r^2}\left(\frac{r^3-r_{min}^3}{r_{max}^3-r_{min}^3}\right)}$$

Thus,



Problem 15) known: A: cross-sectional area   ρ: Earth uniform density
R: Earth radius

unknown: Force on the entire tube?

Consider a tiny element of mass in the tube at distance r from Earth center. Only the mass inside radius of r is relevant. Thus,

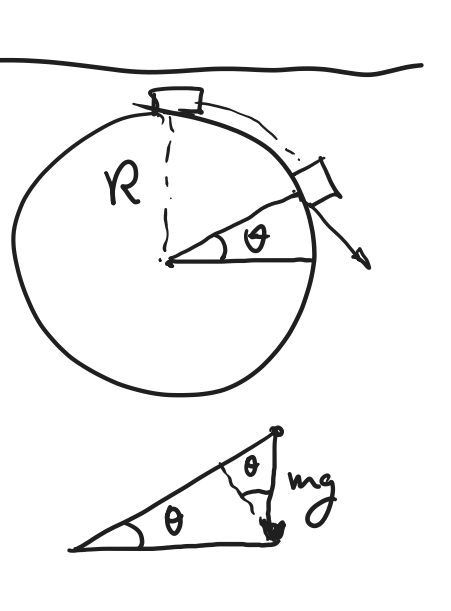
$$dF(r)=\frac{GM(r'<r)dm}{r^2}=\frac{Gdm}{r^2}\left(\frac{4}{3}\pi r^3\rho\right)=\frac{4}{3}\pi G\rho r\,dm$$

Note also, $dm=\rho A dr$

$$\Rightarrow\int_0^{F(R)}dF(r)=\int_0^R\frac{4}{3}\pi G\rho^2 rA\,dr=\frac{2}{3}\pi G\rho^2Ar^2\Big|_0^R$$

$$\Rightarrow\boxed{F(R)=\frac{2}{3}\pi G\rho^2AR^2}$$

Problem 16) The detachment occurs where the inward component of gravitational force becomes insufficient to represent the centripetal force. Let this point's angle be θ.



$\Rightarrow F_c=\frac{mv^2}{R}=mg\sin\theta$ Ⓘ two unknowns. But we also know from work-energy theorem that the vertical distance from the top to the detachment point is $R-R\sin\theta$.

$$\left.\begin{aligned}\frac{1}{2}mv^2=mgR(1-\sin\theta)\quad\text{(II)}\\\text{Ⓘ}\end{aligned}\right\}\Rightarrow 2mg(1-\sin\theta)=mg\sin\theta$$

$$\Rightarrow 3\sin\theta=2\Rightarrow\sin\theta=2/3\Rightarrow\boxed{\theta\simeq 0.7297\text{ rad}}$$

Problem 17) To escape to infinity we must start with a velocity that minimally corresponds to the same kinetic energy as potential energy.

$$T+U=0\Rightarrow\frac{1}{2}mv^2=\frac{GMm}{R}\Rightarrow\boxed{v=\sqrt{\frac{2GM}{R}}}$$

$$v_{Earth}\simeq 11\text{ km/s}\ ;\ v_{moon}\simeq 2.3\text{ km/s}\ ;\ v_{sun}\simeq 620\text{ km/s}$$

Problem 18) The work-energy theorem states that the initial release velocity would be $v_0=\sqrt{2gl\cos\theta}$ where l represents the string length. From projectile motion in Newtonian dynamics, the horizontal distance traveled is: $x=\frac{2v_0^2\sin\theta\cos\theta}{g}=\frac{2(2gl\cos\theta)\sin\theta\cos\theta}{g}$

$$\frac{dx}{d\theta}=\cos^3\theta-2\cos\theta\sin^2\theta=0\Rightarrow\cos^2\theta=2\sin^2\theta\Rightarrow\tan\theta=\frac{1}{\sqrt{2}}$$